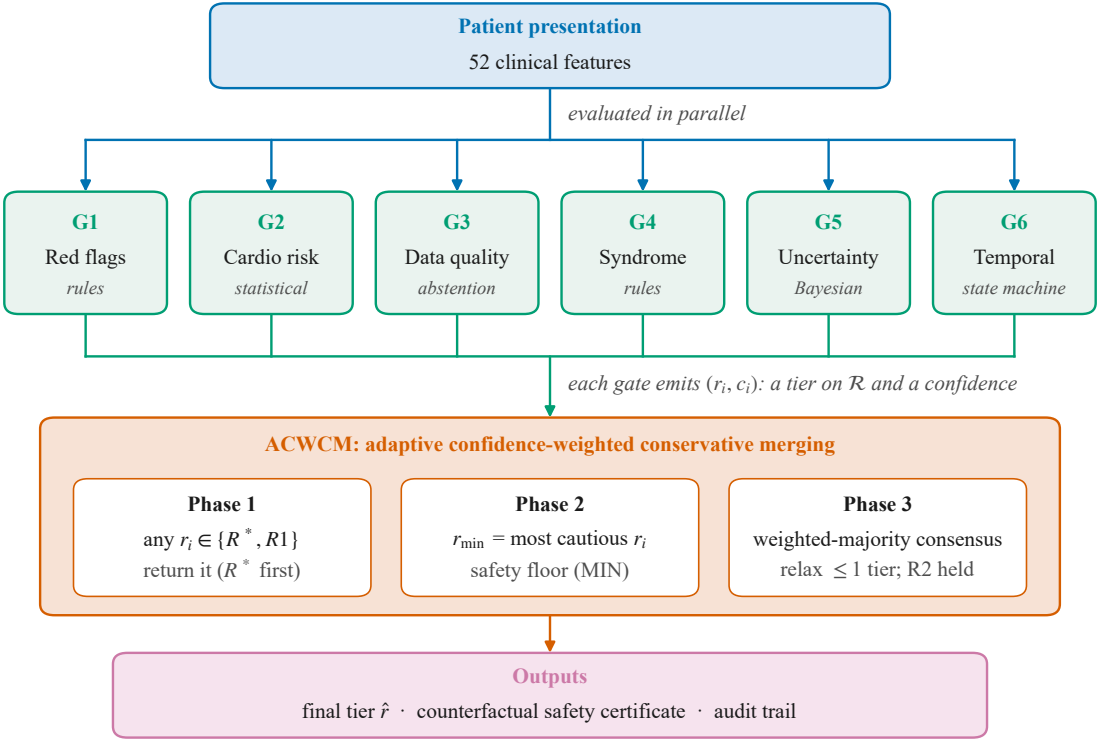

Patient presentation
52 clinical features
evaluated in parallel
G1
Red flags
rules
G2
Cardio risk
statistical
G3
Data quality
abstention
G4
Syndrome
rules
G5
Uncertainty
Bayesian
G6
Temporal
state machine
each gate emits (r_i, c_i): a tier on $\mathcal{R}$ and a confidence
ACWCM: adaptive confidence-weighted conservative merging
Phase 1
any $r_i \in \{R^*, R1\}$
return it (R^* first)
Phase 2
$r_{\min}$ = most cautious r_i
safety floor (MIN)
Phase 3
weighted-majority consensus
relax ≤ 1 tier; R2 held
Outputs
final tier $\hat{r}$ · counterfactual safety certificate · audit trail